Climate Informatics 2026 (2026), xx: 1–2
doi:10.1017/xxxx

CAMBRIDGE UNIVERSITY PRESS

CLIMATE INFORMATICS 2026
EXTENDED ABSTRACT

A Spectral & Causal Analysis Method for Air–Sea Coupling Feedbacks in Eddy-Resolving Earth System Models

Aaron Wienkers[1] and Nicolas Gruber[1]

[1]Department of Environmental System Sciences, ETH Zurich, Zürich, Switzerland

Keywords: air–sea coupling, causal machine learning, process-level evaluation, AI weather models, mesoscale dynamics

1. Motivation

Air–sea thermal & current feedbacks govern energy and momentum exchange with fundamentally different characteristics across spatial scales. These scale-dependent feedbacks regulate ocean heat transport and circulation patterns critical to climate model fidelity. Thermal air–sea feedback exhibits competing pathways: large-scale (>300 km) thermodynamic instability via the "Wind–Evaporation–SST" (WES) mechanism versus mesoscale (10–250 km) dynamic damping via the "Vertical Mixing Mechanism" (VMM) (Small et al., 2008). The current feedback shows analogous scale dependence. Traditional coupling coefficients conflate these distinct mechanisms within correlation-based metrics, obscuring scale-dependent processes and masking compensating errors between competing pathways.

These correlation-based diagnostics likewise cannot establish causality nor distinguish whether models (including AI weather/climate models) achieve realistic air–sea interactions by representing the fundamental physical processes or by directly fitting to emergent behaviour. Causal methods have not yet been rigorously applied to characterise the directionality of air–sea coupling across spatial scales, leaving open whether models reproduce the correct answer for the right reason (Nowack et al., 2020).

We introduce a spatially-resolved framework combining nonlinear spatial-scale decomposition with the PCMCI+ causal discovery algorithm (Runge, 2020) to disentangle feedback mechanisms, enabling process-level evaluation of whether models reproduce physically correct causal pathways alongside realistic coupling magnitudes.

2. Methods

Daily sea surface temperature T_o and 10-metre wind speed $|\mathbf{u}_a|$ from the EERIE km-scale global coupled ICON simulation ($\sim$5 km resolution, 10 years) are nonlinearly decomposed into mesoscale and large-scale components via a spatial filter with a locally-adaptive Rossby-radius lengthscale $\mathcal{L}_{\mathrm{Ro}}$. Fields are then deseasonalised and detrended to satisfy PCMCI+ stationarity requirements.

PCMCI+ (Runge, 2020) is applied independently at each grid cell to T_o and $|\mathbf{u}_a|$, for both the mesoscale and large-scale components. PCMCI+ substantially outperforms lagged-correlation and Granger causality for autocorrelated geophysical fields by conditioning on a selected set of autoregressive parents of both variables (via the PC algorithm) rather than their full lag history. It additionally detects contemporaneous ($\tau = 0$) causal links, which are important for capturing fast boundary layer adjustments that appear instantaneous at daily resolution. The MCI test returns partial-correlation effect sizes $\hat{\rho}^{\mathrm{MCI}}$ for each directional link, which are then spatially aggregated into $1^\circ \times 1^\circ$ bins.

© The Authors(s) 2020. Published by Cambridge University Press. This is an Open Access article, distributed under the terms of the Creative Commons Attribution licence (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted re-use, distribution, and reproduction in any medium, provided the original work is properly cited.

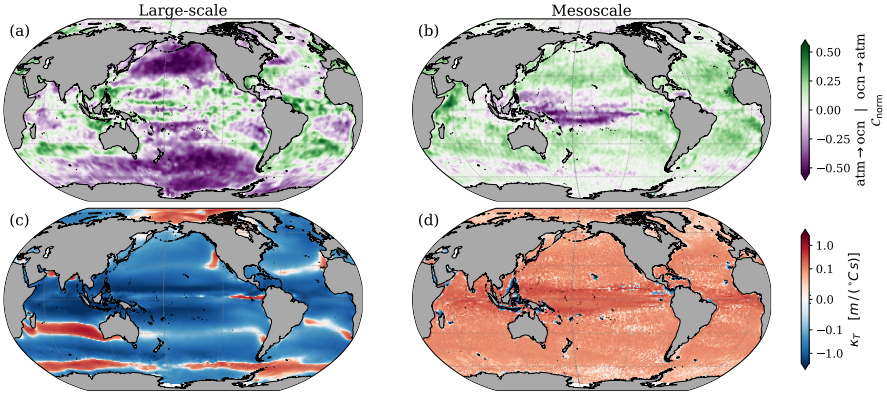

Figure 1. *Global maps comparing the thermal feedback coupling coefficient* $\kappa_T \equiv T_o \wedge |\mathbf{u}_a|$ *(c,d) with the normalised causal arrow* $\mathcal{C}_{\mathrm{norm}}$ *(a,b) at the mesoscale (*$< 20\,\mathcal{L}_{\mathrm{Ro}}$*; b,d) and large-scale (a,c).* $\mathcal{C}_{\mathrm{norm}} > 0$ *indicates dominant ocean-to-atmosphere causality and vice versa for* $\mathcal{C}_{\mathrm{norm}} < 0$*.*

We synthesise the bidirectional MCI effect sizes into a normalised causal arrow,

$$\mathcal{C}_{\mathrm{norm}} = \frac{\left|\hat{\rho}^{\mathrm{MCI}}_{T_o \to |\mathbf{u}_a|}\right| - \left|\hat{\rho}^{\mathrm{MCI}}_{|\mathbf{u}_a| \to T_o}\right|}{\left|\hat{\rho}^{\mathrm{MCI}}_{T_o \to |\mathbf{u}_a|}\right| + \left|\hat{\rho}^{\mathrm{MCI}}_{|\mathbf{u}_a| \to T_o}\right|} \in [-1,\, 1], \tag{1}$$

where $\mathcal{C}_{\mathrm{norm}} = +1$ denotes complete ocean-to-atmosphere causality and $\mathcal{C}_{\mathrm{norm}} = -1$ atmosphere-to-ocean. The traditional thermal coupling coefficient, $\kappa_T \equiv T_o \wedge |\mathbf{u}_a|$ (where $X \wedge Y$ denotes the slope of the binned robust linear regression of Y on X), provides a baseline for comparison with $\mathcal{C}_{\mathrm{norm}}$.

3. Results

Scale-decomposed PCMCI+ reveals physically distinct causal structures at each scale (Figure 1). At mesoscales, $\kappa_T > 0$ and $\mathcal{C}_{\mathrm{norm}} > 0$ globally, consistent with VMM ($T_o \uparrow \implies |\mathbf{u}_a| \uparrow$). Two regions depart from this picture. Over the tropical Pacific, κ_T remains positive yet $\mathcal{C}_{\mathrm{norm}} < 0$: the shallow thermocline means mesoscale wind anomalies affect T_o via wind-driven upwelling and mixed-layer entrainment at daily lags, outweighing the lagged VMM signal. Intense storm tracks over the Southern Ocean similarly lead to mesoscale wind anomalies that modify T_o through turbulent heat fluxes and Ekman upwelling. In both cases, the atmosphere-dominant causality ($\mathcal{C}_{\mathrm{norm}} < 0$) is masked by $\kappa_T > 0$.

At large scales, $\kappa_T < 0$ broadly, consistent with WES and stochastic atmospheric forcing, but with notable exceptions over strong upwelling regions. $\mathcal{C}_{\mathrm{norm}}$ identifies atmospherically forced dynamics poleward of $\pm 60°$, even where $\kappa_T > 0$ would suggest the atmosphere-driven WES is not active. At lower latitudes, the spatially heterogeneous $\mathcal{C}_{\mathrm{norm}}$ suggests that multiple large-scale processes, including Bjerknes feedback, MJO, & WES, collectively shape the large-scale causal structure.

We finally apply this new evaluation framework to the EERIE km-scale multi-model ensemble and AI (neural) emulators to evaluate how physics- & data-driven models reproduce these causal pathways.

Acknowledgements. This work was supported by the EU Horizon Europe Grant 101081383 through the EERIE project.

References

Nowack, P., et al. (2020). Causal networks for climate model evaluation and constrained projections. *Nature Comms.* **11**, 1415.

Runge, J. (2020). Discovering contemporaneous and lagged causal relations in autocorrelated nonlinear time series datasets. *Proceedings of Machine Learning Research* **124**, 1388–1397.

Small, R. J., et al. (2008). Air-sea interaction over ocean fronts and eddies. *Dynamics of Atmospheres and Oceans* **45**, 274–319.